Fiscal Year:	FY 2017	**Task Last Updated:**	FY 10/01/2018
PI Name:	Delp, Michael Ph.D.		
Project Title:	Disuse Osteopenia: A Potential Vascular Coupling Mechanism		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
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No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2017 per F. Hernandez/ARC (Ed., 4/6/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Allen, Matthew Ph.D. (Indiana University School of Medicine) Delp, Judy Ph.D. (Florida State University)		
Grant/Contract No.:	NNX14AQ57G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

NOTE: Continuation of "Disuse Osteopenia: A Potential Vascular Coupling Mechanism," grant #NNX12AL41G. Disuse osteopenia, which is associated with habitation in a microgravity environment, is a condition of low bone density resulting in increased fragility and greater likelihood for bone fracture. Although this condition is largely attributed to the direct effects of unloading of the bone, coupling mechanisms associated with diminished bone and marrow perfusion, attenuated fluid flow within the bone, and fewer signaling molecules from vascular cells may also play a modulating role on the bone loss associated with spaceflight.

In the tail-suspended hindlimb-unloaded (HU) rat, an animal model of microgravity-induced osteopenia, hindlimb bone and marrow blood flow decrease with acute unloading (10 min) and continue to progressively decline through 28 days of unloading. The initial and chronic decrements in perfusion are primarily due to increases in osseous vascular resistance. In addition, when the HU rat hindlimb skeleton is reloaded during standing, long bone and marrow perfusion are lower and vascular resistance higher than that in standing control animals. These higher vascular resistances in HU rats could result from 1) increased vasoconstrictor responses of bone resistance arteries, or 2) decreased endothelium-dependent and/or endothelium-independent vasodilator responsiveness of bone resistance arteries.

Specific Aims:

Aim 1. To determine whether hindlimb unloading alters the responsiveness of the principal nutrient artery (PNA) to myogenic and vasoconstrictor stimuli.

Aim 2. To evaluate the contribution of endothelium-derived NO (nitric oxide) and vasodilator prostaglandins to acetylcholine (ACh), bradykinin (BK), and flow-induced vasodilation in the femoral PNA from control and HU rats.

Aim 3. To determine whether hindlimb unloading alters the expression of endothelial NOS (eNOS), COX-1 and COX-2 mRNA and protein, as well as M2-receptor (ACh) and B2-receptor (BK) mRNA, in the femoral PNA.

Aim 4. To determine whether hindlimb unloading alters the PNA responsiveness to endothelium-independent vasodilator stimuli.

Aim 5. To determine whether hindlimb unloading, radiation exposure, and the combined effects of these treatments alter endothelium-dependent vasodilation through NO or PGI2 signaling pathways, and to determine whether this is related to changes in bone volume and indices of cancellous bone micro-architecture.

The general hypothesis of this proposal is that skeletal unloading induces alterations in skeletal resistance artery function; these alterations serve to diminish bone perfusion and ultimately disrupt mechanisms for maintaining the normal balance between bone formation and bone resorption. The specific purpose is to elucidate vasoconstrictor and vasodilator mechanisms involved in the control of bone and marrow perfusion, including endothelial signaling pathways forming nitric oxide and prostacyclin, in bone during conditions of unloading, and to determine whether deficits in bone blood flow and endothelium-dependent vascular signaling are related to unloading-induced bone loss. To accomplish these goals, the principal nutrient artery of the femur from control and 14 day HU rats will be isolated and cannulated for in vitro experimentation. Mechanisms of smooth muscle and endothelial cell function in the osseous circulation will be determined, as well as how skeletal unloading alters these vasoconstrictor and vasodilator control mechanisms (Aims 1-4). Studies will then be conducted using adenoviral gene transfer (Aim 5) to manipulate bone and marrow perfusion and endothelium-dependent vasodilation to determine whether blood flow and vascular endothelial cell signaling is coupled to disuse-associated bone remodeling in the femur (i.e., bone formation rates, mass, total, cortical and cancellous bone mineral density, and mechanical strength). These studies will extend our knowledge and understanding of how skeletal unloading alters fluid and vascular coupling mechanisms, which have been suggested to modulate the balance of bone formation and degradation during microgravity. These studies may also provide new insight into the etiology of other degenerative bone disorders that occur concomitantly with cardiovascular dysfunction, such as heart disease, atherosclerosis, diabetes, and old age.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

These studies will extend our knowledge and understanding of how skeletal unloading alters fluid and vascular coupling mechanisms, which have been suggested to modulate the balance of bone formation and degradation during microgravity. These studies may also provide new insight into the etiology of other degenerative bone disorders that occur concomitantly with cardiovascular dysfunction, such as heart disease, atherosclerosis, diabetes, and old age.

Task Progress:

[Ed. note Sep 2018: compiled from final report dated spring 2018]
In total, five manuscripts have been published with support derived from this grant (see Cumulative Bibliography below) that address Aims 1-5. Two additional manuscripts that will cite support from this grant are in preparation. In the first unpublished study, we examined whether simulated weightlessness and space-relevant irradiation of mice alters vascular responsiveness and bone properties 6-7 months after these treatment interventions, i.e., following a 6-7 period of normal weight bearing recovery. This period of recovery in mice is equivalent to approximately 20 human years. The data demonstrate that only high-LET (linear energy transfer) total body irradiation (TBI), but not hindlimb unloading (HU), has a sustained effect on endothelium-dependent vasodilation following the period of recovery, and that the combined effects of these two treatments (TBI-HU) to impair endothelium-dependent vasodilation is also sustained following the recovery period. The was a tendency for the trabecular bone volume fraction (trabecular bone volume/total bone volume) of the proximal tibia from TBI-HU mice to be lower than that in control mice, but this was not significant. Decrements in endothelium-dependent vasodilation are significantly correlated with changes in bone volume, indicating a coupling of vascular endothelial cell function is associated with bone loss induced by HU and TBI. This observation is similar to that which we reported earlier in mice where the effects of HU and TBI were determined immediately after the cessation of these treatments (Ghosh et al., Radiat. Res. 185: 257-266, 2016), although the relation between peak endothelium-dependent vasodilation and bone volume fraction was not as robust following the recovery period. Collectively, these data demonstrate the important relation between bone blood flow, bone vascular function, and trabecular bone volume.

For the second manuscript in preparation, we examined the effects of a commonly used countermeasure for spaceflight-induced bone loss (i.e., exercise) to determine its effect on bone blood flow. Like that shown to occur in hindlimb bones with chronic unloading (Stabley et al., 2013), old age also diminishes perfusion of bone marrow and regions of bone containing high volumes of trabecular bone. Results from this study demonstrate that 10-12 weeks of

	low-intensity exercise training (ET) restores bone perfusion in old (25-26 months old) rats to levels occurring in young adult (6 months old) animals. Thus, these data support the notion that regular physical activity maintains and improves bone perfusion and bone health. Stabley JN, Prisby RD, Behnke BJ, Delp MD. "Chronic skeletal unloading of the rat femur: mechanisms and functional consequences of vascular remodeling." Bone. 2013 Dec;57(2):355-60.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Articles in Peer-reviewed Journals	Muller-Delp JM, Hotta K, Chen B, Behnke BJ, Maraj JJ, Delp MD, Lucero TR, Bramy JA, Alarcon DB, Morgan HE, Cowan MR, Haynes AD. "Effects of age and exercise training on coronary microvascular smooth muscle phenotype and function." J Appl Physiol (1985). 2018 Jan 1;124(1):140-9. Epub 2017 Oct 12. https://doi.org/10.1152/japplphysiol.00459.2017 ; PubMed PMID: 29025901; PubMed Central PMCID: PMC5866446 , Jan-2018
Articles in Peer-reviewed Journals	Stabley JN, Prisby RD, Behnke BJ, Delp MD. "Type 2 diabetes alters bone and marrow blood flow and vascular control mechanisms in the ZDF rat." J Endocrinol. 2015 Apr;225(1):47-58. http://dx.doi.org/10.1530/JOE-14-0514 ; PubMed PMID: 25817711; PubMed Central PMCID: PMC4379453 , Apr-2015
Articles in Peer-reviewed Journals	Ghosh P, Mora Solis FR, Dominguez JM 2nd, Spier SA, Donato AJ, Delp MD, Muller-Delp JM. "Exercise training reverses aging-induced impairment of myogenic constriction in skeletal muscle arterioles." J Appl Physiol (1985). 2015 Apr 1;118(7):904-11. Epub 2015 Jan 29. https://doi.org/10.1152/japplphysiol.00277.2014 ; PubMed PMID: 25634999; PubMed Central PMCID: PMC4422370 , Apr-2015